

Project Design Phase

Problem – Solution Fit Template

Date	10 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Problem – Solution Fit Canvas:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

 1. CUSTOMER SEGMENTS (CS) • Farmers • Agricultural Researchers • Agribusiness Companies • Agricultural Extension Officers • Government Agricultural Departments	 2. CUSTOMER CONSTRAINTS (CC) • Difficulty selecting suitable crops • Lack of scientific recommendations • Unpredictable weather conditions • Limited access to experts • Inefficient resource utilization • Financial losses due to poor crop selection	 3. AVAILABLE SOLUTIONS (AS) • Traditional farming knowledge • Consultation with agricultural experts • Government advisory services • Weather apps • Manual soil testing reports	
 4. JOBS-TO-BE-DONE / PROBLEMS (JB) • Select the most suitable crop • Maximize crop yield • Reduce farming risks • Improve resource utilization • Make data-driven farming decisions	 5. PROBLEM ROOT CAUSE (RC) • Lack of AI-based decision support • Dependence on traditional farming practices • Changing climatic conditions • Improper analysis of soil nutrients and environmental parameters	 6. BEHAVIOUR (BE) • Farmers usually depend on previous experience, neighboring farmers, local agricultural officers, or trial-and-error methods before choosing crops.	
 7. TRIGGERS (TR) • Poor harvest • Financial loss • Changing weather conditions • New cultivation season • Government awareness programs • Adoption of digital agriculture technologies	 8. CHANNELS OF BEHAVIOUR (CH) • Agricultural websites • Mobile applications • Agricultural officers • Farmer communities • Social media • Agricultural training programs	 9. EMOTIONS BEFORE / AFTER (EM) Before: Confused, uncertain, worried about crop failure and income. After: Confident, satisfied, motivated and optimistic after receiving accurate crop recommendations.	 10. YOUR SOLUTION (SL) OptiCrop is an AI-powered crop recommendation system that uses Machine Learning to analyze Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop, enabling data-driven farming decisions and improving agricultural productivity.